EC2L104 – Introduction to EDA Tools

LTspice Exercise Report

MOSFET Characterisation, Parametric Study and CMOS Inverter
using the 65 nm PTM Model Card

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GitHub repo: _____ (`LTSPICE`)

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1 S1 – Reading the Model Card

Circuit / Testbench

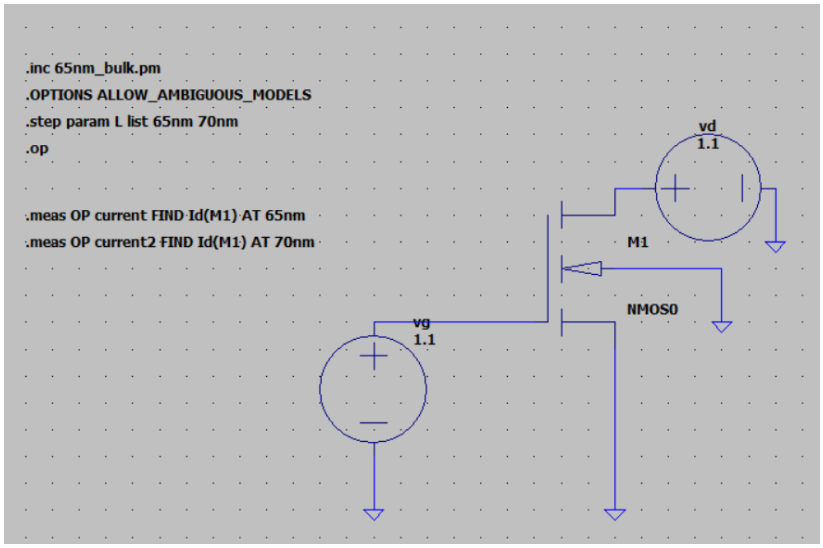


Figure 1: LTspice testbench for the `.op` runs ($L = 65$ nm and 70 nm).

```
.step param L list 65nm 70nm
.op
.meas OP current FIND Id(M1) AT 65nm
.meas OP current2 FIND Id(M1) AT 70nm
```

1.1 Extracted Card Parameters

Parameter	Value
XL (nm)	-30
LINT (nm)	5.25
WINT (nm)	5
TOXE (nm)	1.85

Table 1: Values read from the model card.

1.2 Calculations

$$L_{eff} = L_{drawn} + XL - 2\,LINT = L_{drawn} - 40.5 \text{ nm}$$
$$W_{eff} = W_{drawn} - 2\,WINT = W_{drawn} - 10 \text{ nm}$$
$$C_{ox} = \frac{3.9\,\varepsilon_0}{TOXE} = 1.87\ \mu\text{F/cm}^2$$

L (nm)	L_{eff} (nm)
65	24.5
70	29.5

Table 2: Computed effective channel length.

1.3 .op Results at $V_{GS} = V_{DS} = 1.1$ V

L (nm)	I_D (mA)
65	1.15
70	1.05

Table 3: Drain current from `.op`.

Observations

We can see that with increase in channel length a proportional decrease in drain current.

$$\frac{L_{70}}{L_{65}} = 1.08$$

$$\frac{I_{D65}}{I_{D70}} = \frac{1.15}{1.05} = 1.09$$

$$\frac{L_{70}}{L_{65}} \approx \frac{I_{D65}}{I_{D70}}$$

Conclusion

From the observation we can conclude that I_D **varies inversly proportional to channel length** which matches to the theory we learnt.

$$I_D \propto \frac{1}{L}$$

2 S2 – Transfer Characteristics

Circuit / Testbench

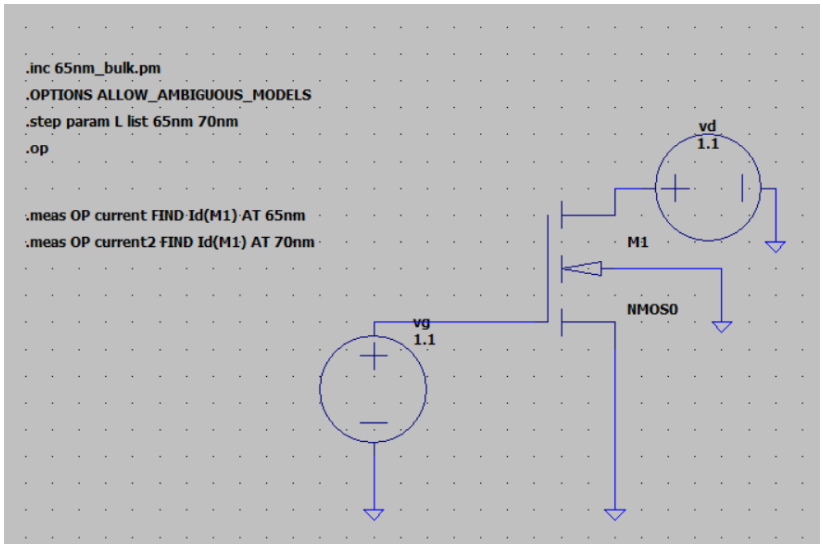


Figure 2: LTspice testbench for I_D–V_G (V_{DS} = 50 mV and 1.1 V).

```
.dc vg 0 1.1 0.01
.step param v list 0.05 1.1
```

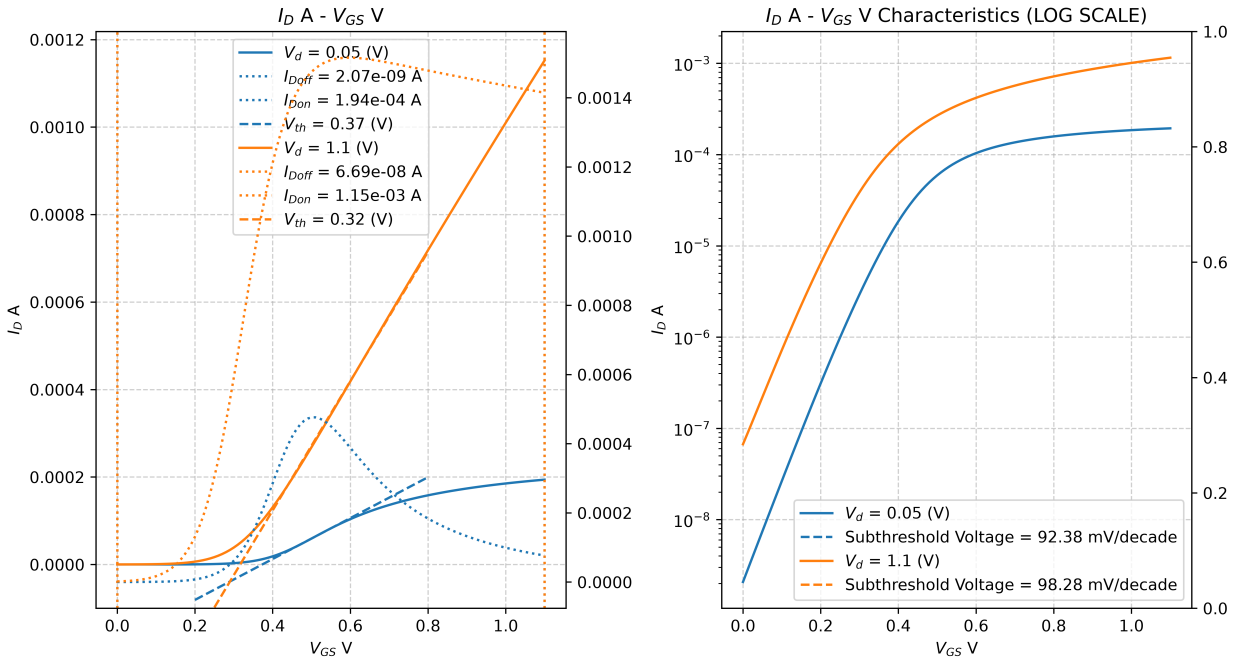


Figure 3: I_D–V_G at V_{DS} = 50 mV and 1.1 V (L = 65 nm).

Parameter	Method	Extracted (SPICE)	Table 5.1
V_{TH} (V)	Tangent at maximum gm, VDS = 50 mV, intercept with I_D =0	0.37	0.38
I_{ON} (mA/μm)	$V_{GS} = V_{DS} = 1.1$ V	1.15	1.15
I_{OFF} (nA/μm)	$V_{GS} = 0$, $V_{DS} = 1.1$ V	66.9	67
SS (mV/dec)	Extrapolation similar to V_{th}	92.38 (0.05 V), 98.28 (1.1 V),	90 (50 mV), 96(1.1 V)

Table 4: Comparison of extracted parameters

3 S3 – Output Characteristics

Circuit / Testbench

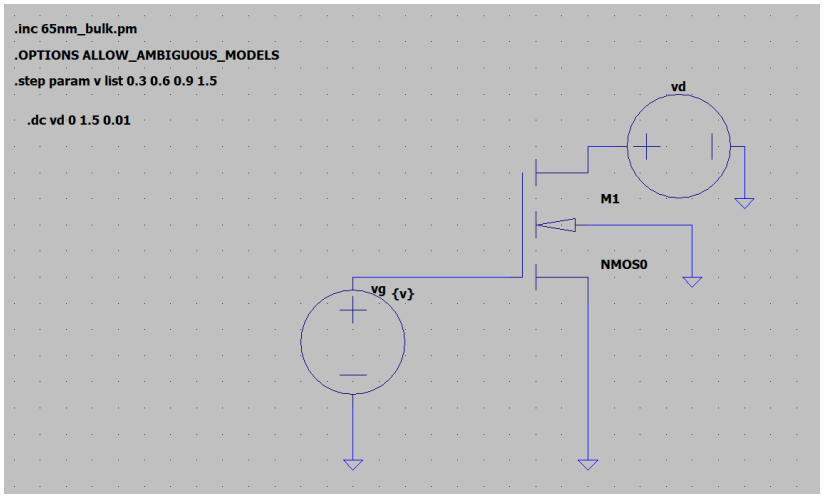


Figure 4: LTspice testbench for the I_D–V_D family.

```
.dc vd 0 1.5 0.01
.step param v list 0.3 0.6 0.9 1.5
```

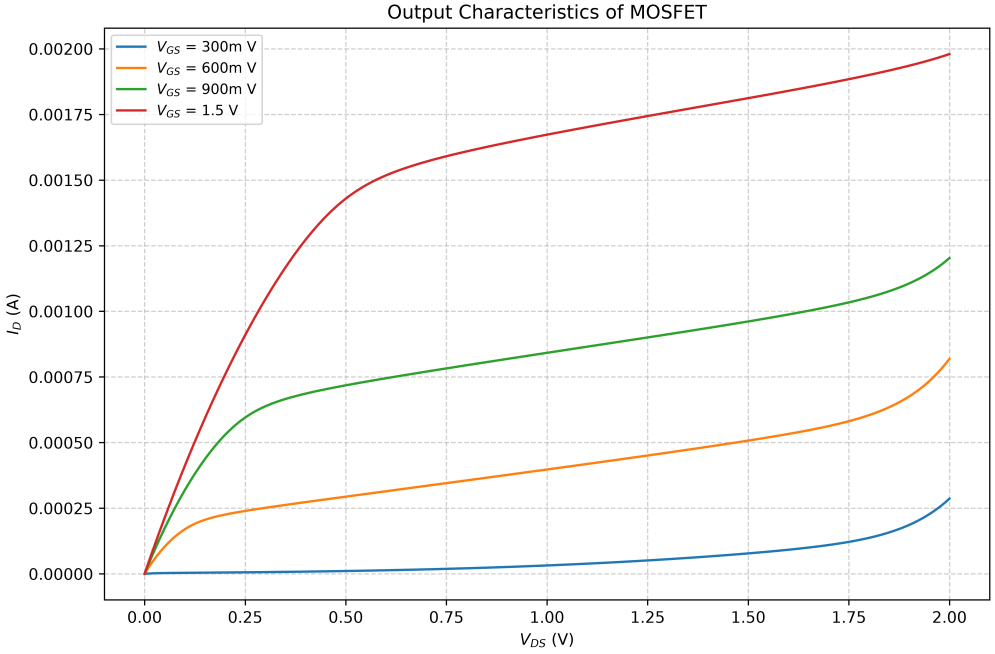


Figure 5: I_D–V_D family at $V_{GS} = 0.3$, 0.6, 0.9, 1.5 V.

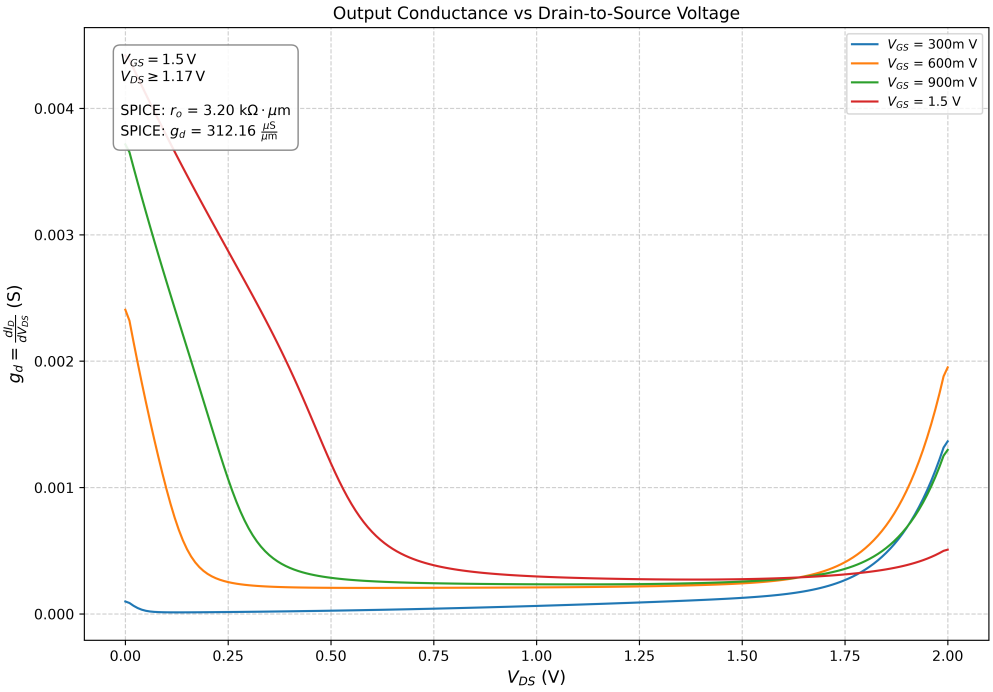


Figure 6: g_{ds} (via `np.gradient`) versus V_{DS}; saturation region marked at $V_{GS} = 1.5$ V.

Quantity (at $V_{GS} = 1.5$ V, saturation)	SPICE
V_{DS} range used for g_{ds} (V)	1.17 - 2
g_{ds} (μS/μm)	312.16
$r_o = 1/g_{ds}$ (k$\Omega \cdot \mu$m)	3.20

Table 5: Output conductance and resistance.

4 S4 – Physics Check

Circuit / Testbench

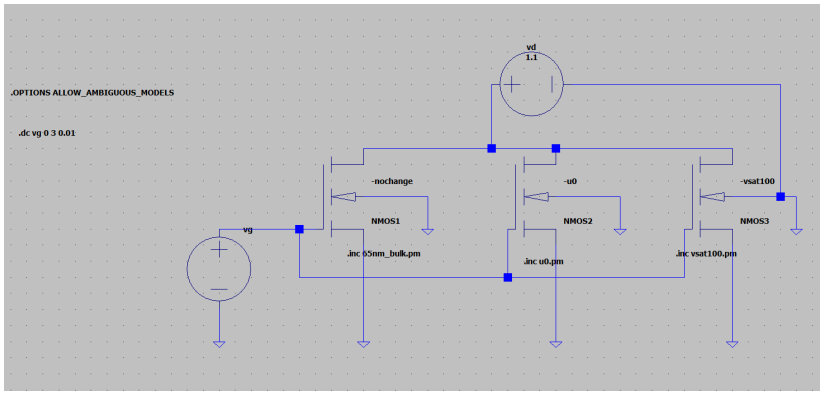


Figure 7: LTspice testbench for the model-parameter checks (UA, UB, $VSAT$).

```
.dc vg 0 3 0.01
```

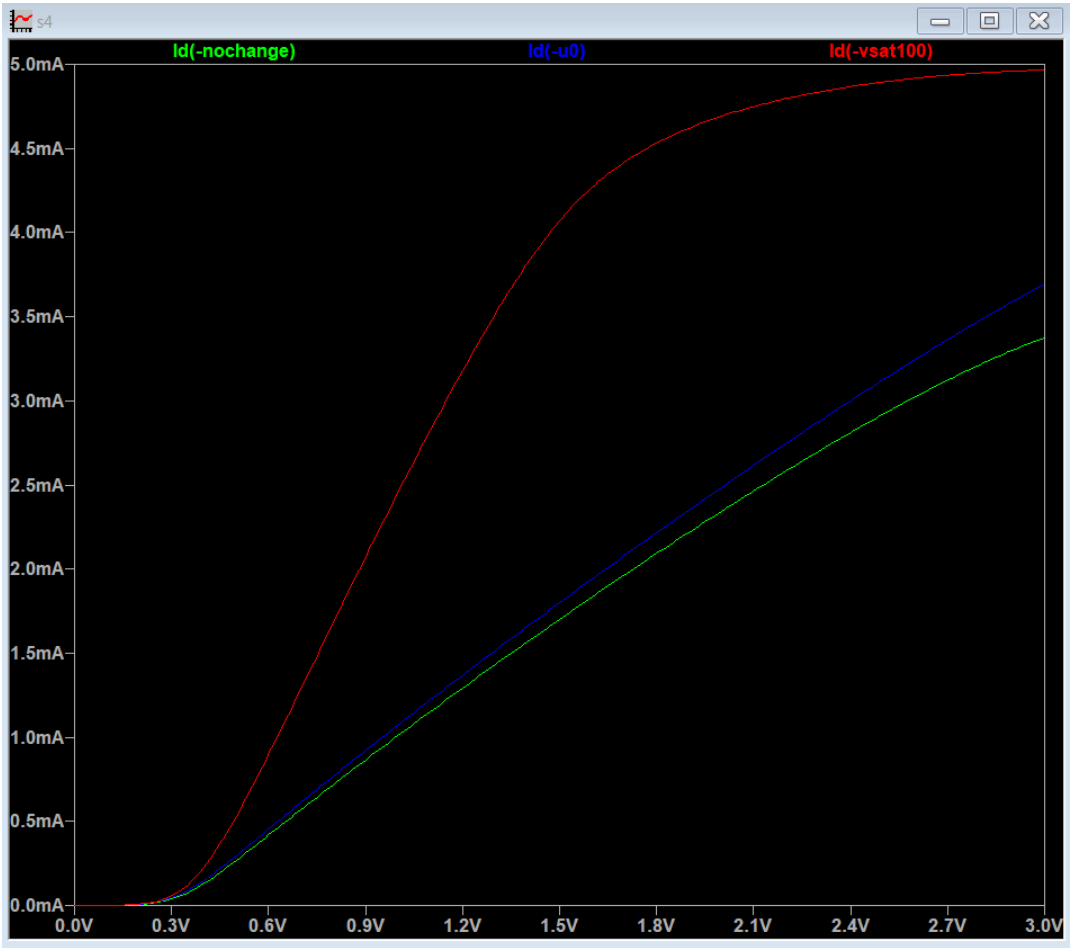


Figure 8: I_D–V_G overlay: nominal, $UA = UB = 0$, and $VSAT \times 100$.

Device-physics explanation

$UA = UB = 0 \leftrightarrow$ **TCAD *Enormal* (mobility degradation) model:**
$VSAT \times 100 \leftrightarrow$ **TCAD *HighFieldSaturation* model:** *[explain removal of velocity saturation, effect on I_{ON} and saturation behaviour.]*

Conclusion

[Which effect limits I_{ON} most in a 65 nm device?]

5 S5 – Parametric Variation

Circuit / Testbench

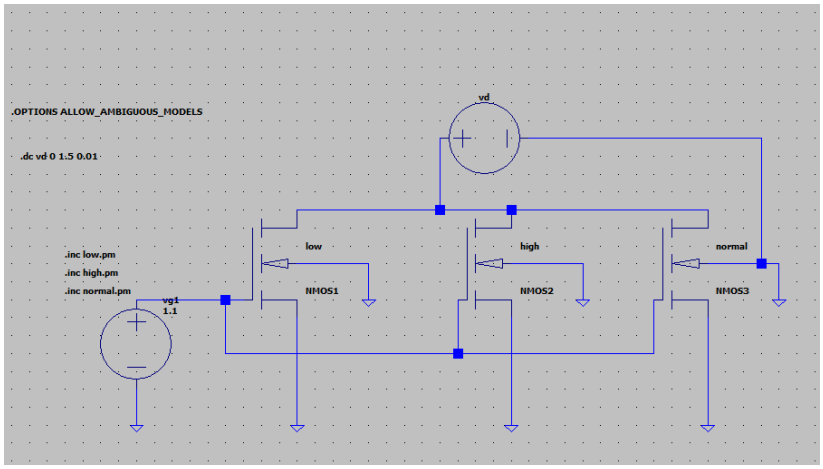


Figure 9: Circuit of I_D-V_G

```
.dc vg 0 1.5 0.01
```

Circuit / Testbench

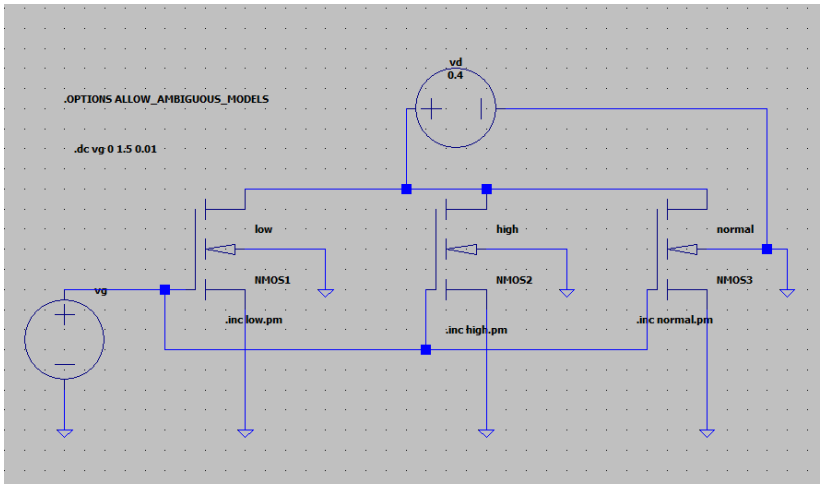


Figure 10: Circuit of I_D-V_D

```
.dc vd 0 1.5 0.01
```

Each experiment varies one parameter around its nominal value. I_D–V_G is simulated at $V_{DS} = 0.4$ V and I_D–V_D at $V_{GS} = 1.1$ V.

Summary of Experiments

Exp.	Parameter	Low / Nominal / High	Unit
1	VTH0	0.30 / 0.423 / 0.55	V
2	TOXE	1.5 / 1.85 / 2.5	nm
3	L (drawn)	55 / 65 / 100	nm
4	XL	-35 / -30 / -15	nm
5	LINT	2 / 5.25 / 8	nm
6	W (drawn)	0.5 / 1 / 2	μm
7	WINT	0 / 5 / 40	nm
8	NDEP	1.5×10^{18} / 2.54×10^{18} / 4×10^{18}	cm^{-3}
9	NSD	5×10^{19} / 2×10^{20} / 5×10^{20}	cm^{-3}
10	XJ	10 / 19.6 / 30	nm
11	NGATE	1×10^{19} / 2×10^{20} / 1×10^{21}	cm^{-3}
12	U0	300 / 491 / 700	$\mathrm{cm}^2/\mathrm{V{\cdot}s}$
13	VSAT	0.8×10^{7} / 1.24×10^{7} / 2×10^{7}	cm/s
14	RDSW	50 / 165 / 400	$\Omega\cdot\mu$m

Table 6: Parameters varied in S5 (the handout text says “1 to 13” but lists 14 rows).

5.1 Experiment 1: VTH0

Values (low / nominal / high): 0.30 / 0.423 / 0.55 V

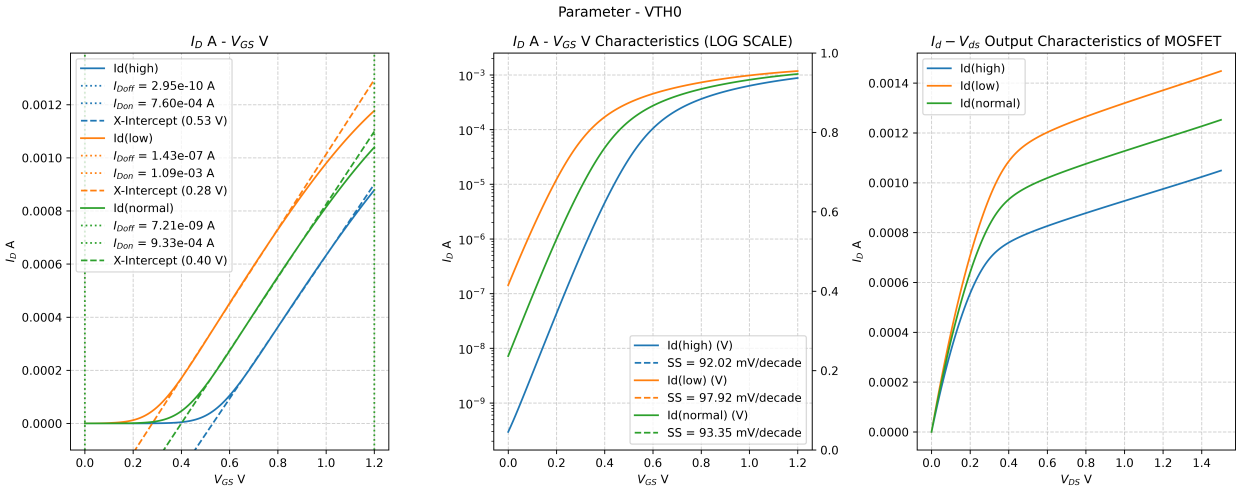


Figure 11: Exp. 1: effect of varying VTH0. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.28	97.92	1090	143
Nominal	0.40	93.35	933	7.21
High	0.53	92.02	760	0.295

Table 7: Extracted parameters, Exp. 1.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.2 Experiment 2: TOXE

Values (low / nominal / high): 1.5 / 1.85 / 2.5 nm

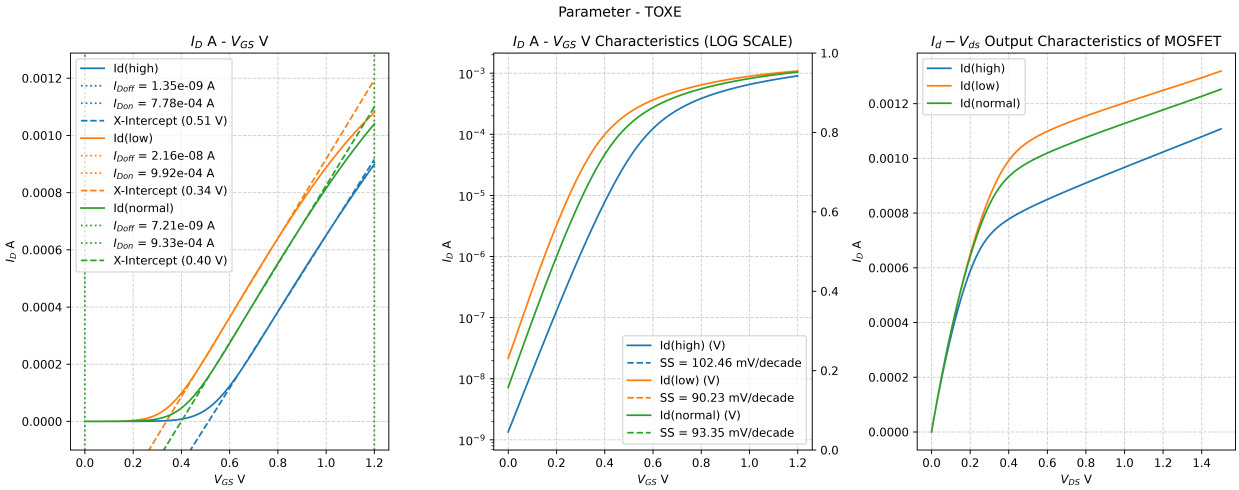


Figure 12: Exp. 2: effect of varying TOXE. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.34	90.23	992	21.6
Nominal	0.40	93.35	933	7.21
High	0.51	102.46	778	1.35

Table 8: Extracted parameters, Exp. 2.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.3 Experiment 3: L (drawn)

Values (low / nominal / high): 55 / 65 / 100 nm

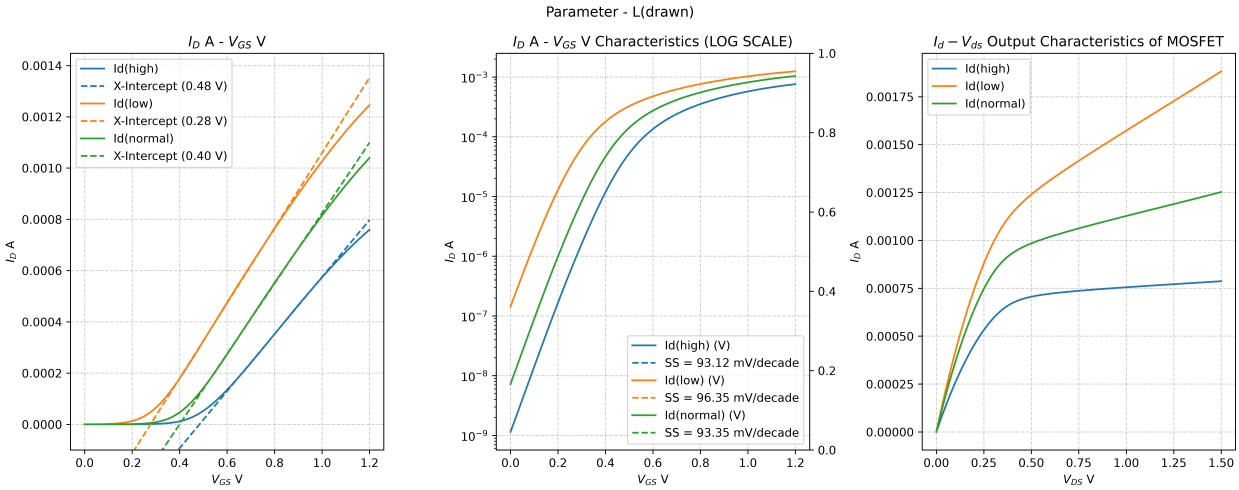


Figure 13: Exp. 3: effect of varying L (drawn). Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.28	96.35	≈1200	≈140
Nominal	0.40	93.35	933	7.21
High	0.48	93.12	≈700	≈1

Table 9: Extracted parameters, Exp. 3.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.4 Experiment 4: XL

Values (low / nominal / high): −35 / −30 / −15 nm

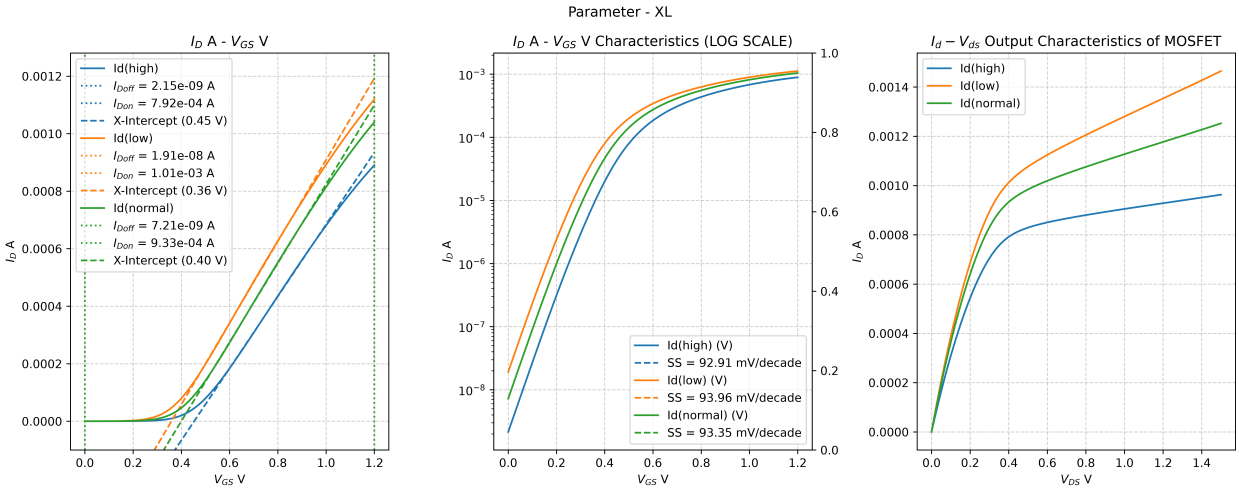


Figure 14: Exp. 4: effect of varying XL. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.36	93.96	1010	19.1
Nominal	0.40	93.35	933	7.21
High	0.45	92.91	792	2.15

Table 10: Extracted parameters, Exp. 4.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.5 Experiment 5: LINT

Values (low / nominal / high): 2 / 5.25 / 8 nm

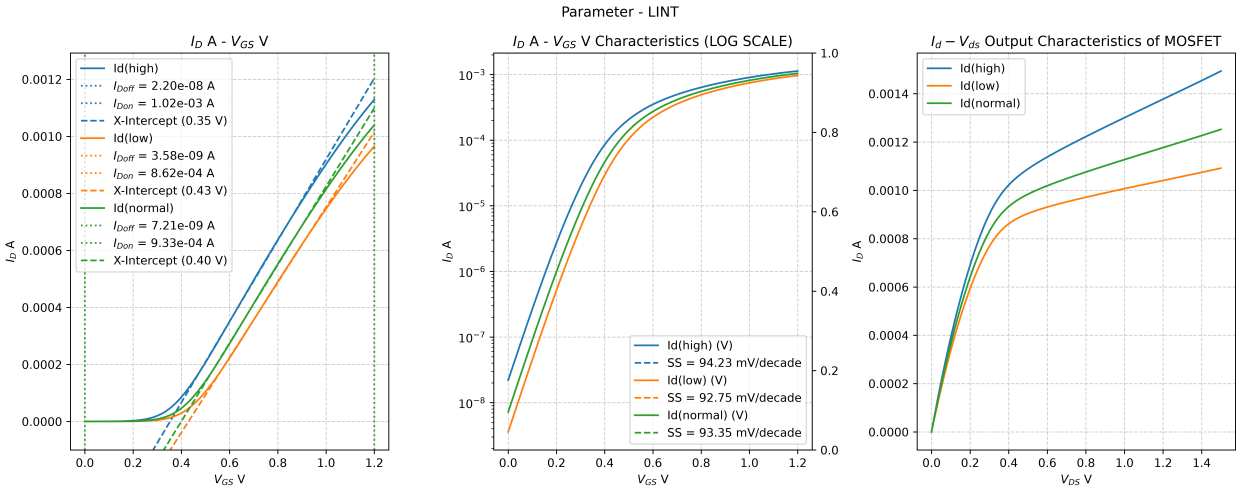


Figure 15: Exp. 5: effect of varying LINT. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.43	92.75	862	3.58
Nominal	0.40	93.35	933	7.21
High	0.35	94.23	1020	22.0

Table 11: Extracted parameters, Exp. 5.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.6 Experiment 6: W (drawn)

Values (low / nominal / high): 0.5 / 1 / 2 μm

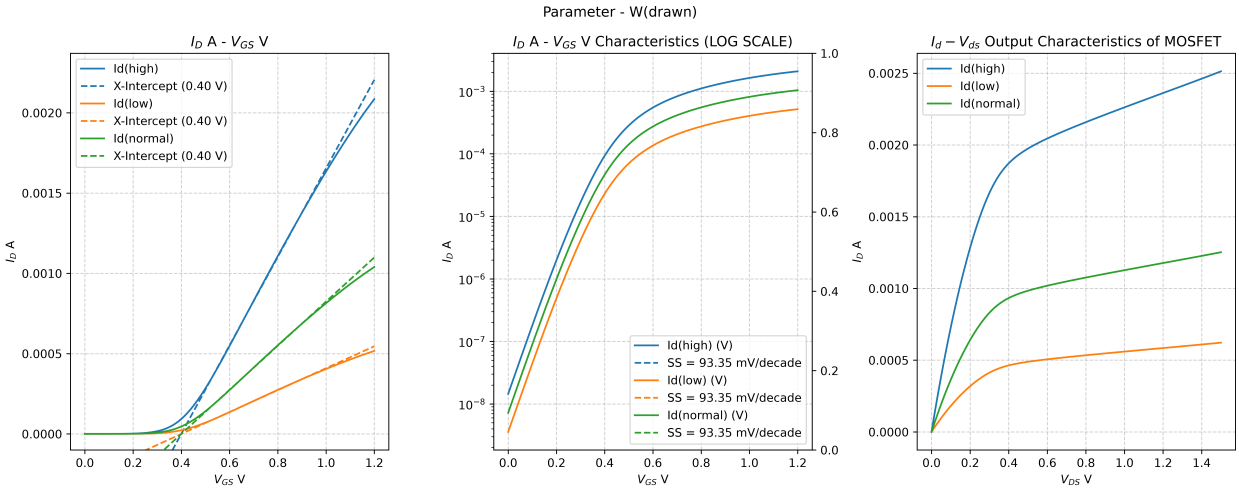


Figure 16: Exp. 6: effect of varying W (drawn). Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.35	933	7.21
Nominal	0.40	93.35	933	7.21
High	0.40	93.35	933	7.21

Table 12: Extracted parameters, Exp. 6.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.7 Experiment 7: WINT

Values (low / nominal / high): 0 / 5 / 40 nm

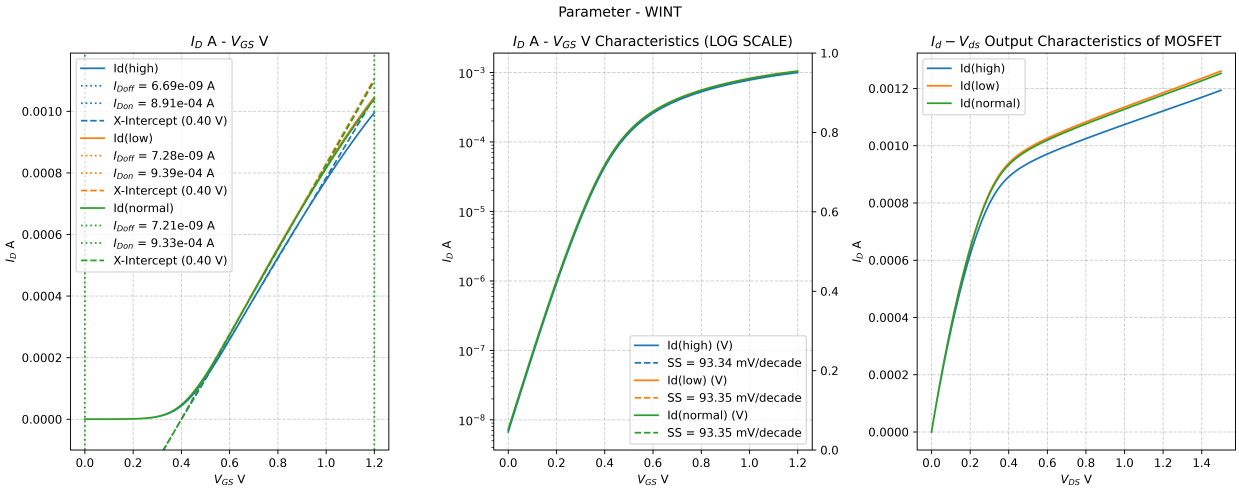


Figure 17: Exp. 7: effect of varying WINT. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.35	939	7.28
Nominal	0.40	93.35	933	7.21
High	0.40	93.34	891	6.69

Table 13: Extracted parameters, Exp. 7.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.8 Experiment 8: NDEP

Values (low / nominal / high): 1.5×10^{18} / 2.54×10^{18} / 4×10^{18} cm^{-3}

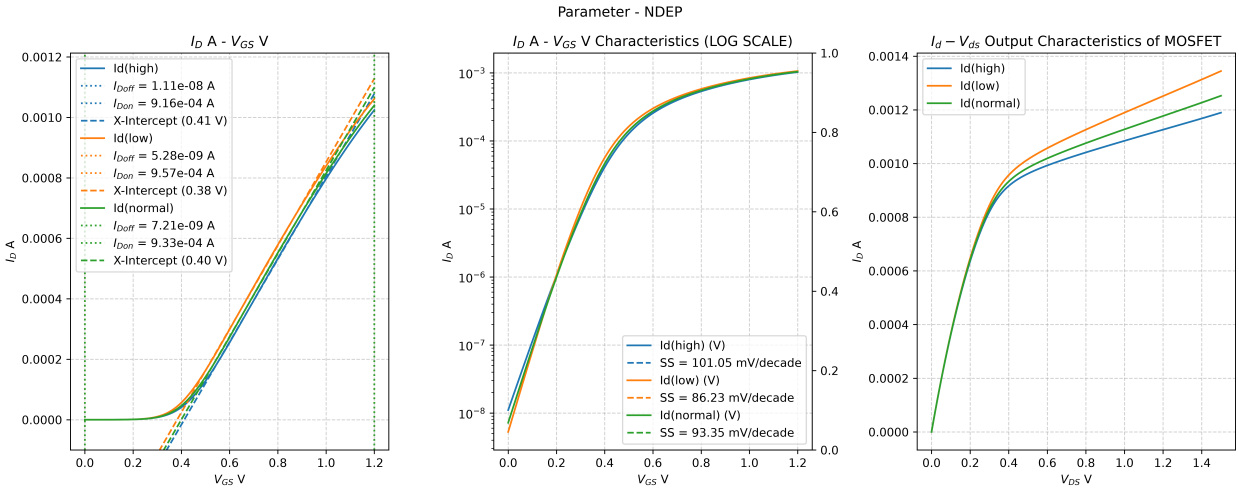


Figure 18: Exp. 8: effect of varying NDEP. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.38	86.23	957	5.28
Nominal	0.40	93.35	933	7.21
High	0.41	101.05	916	11.1

Table 14: Extracted parameters, Exp. 8.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.9 Experiment 9: NSD

Values (low / nominal / high): 5×10^{19} / 2×10^{20} / 5×10^{20} cm^{-3}

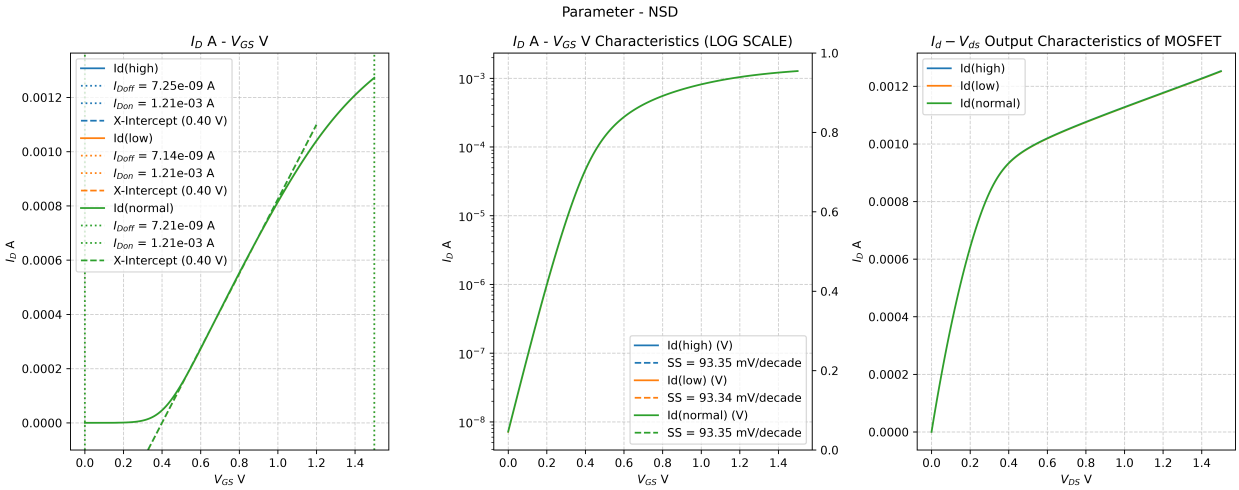


Figure 19: Exp. 9: effect of varying NSD. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.34	1210	7.14
Nominal	0.40	93.35	1210	7.21
High	0.40	93.35	1210	7.25

Table 15: Extracted parameters, Exp. 9.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.10 Experiment 10: XJ

Values (low / nominal / high): 10 / 19.6 / 30 nm

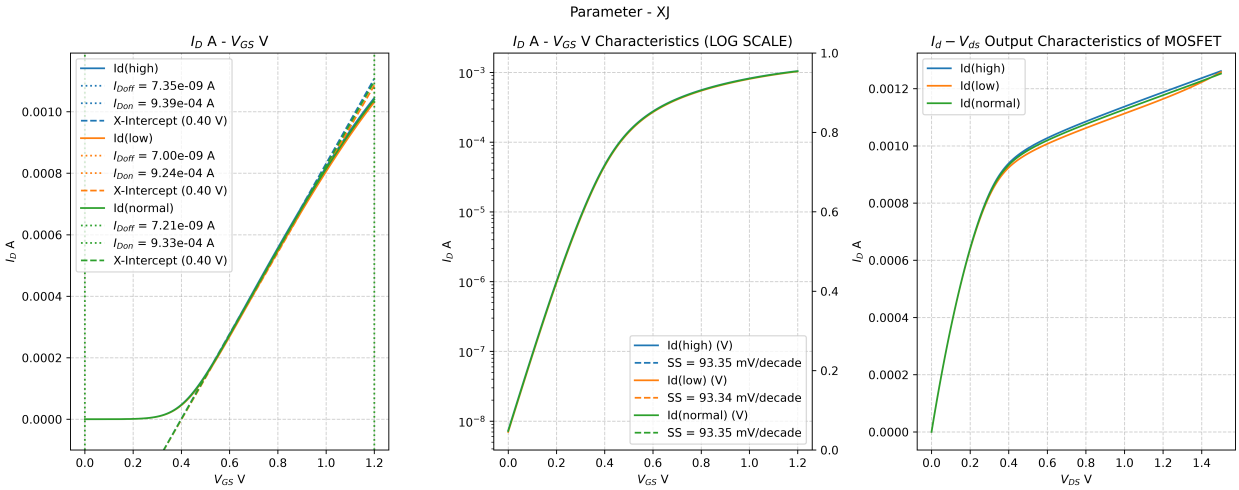


Figure 20: Exp. 10: effect of varying XJ. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.40	93.34	924	7.00
Nominal	0.40	93.35	933	7.21
High	0.40	93.35	939	7.35

Table 16: Extracted parameters, Exp. 10.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.11 Experiment 11: NGATE

Values (low / nominal / high): 1×10^{19} / 2×10^{20} / 1×10^{21} cm^{-3}

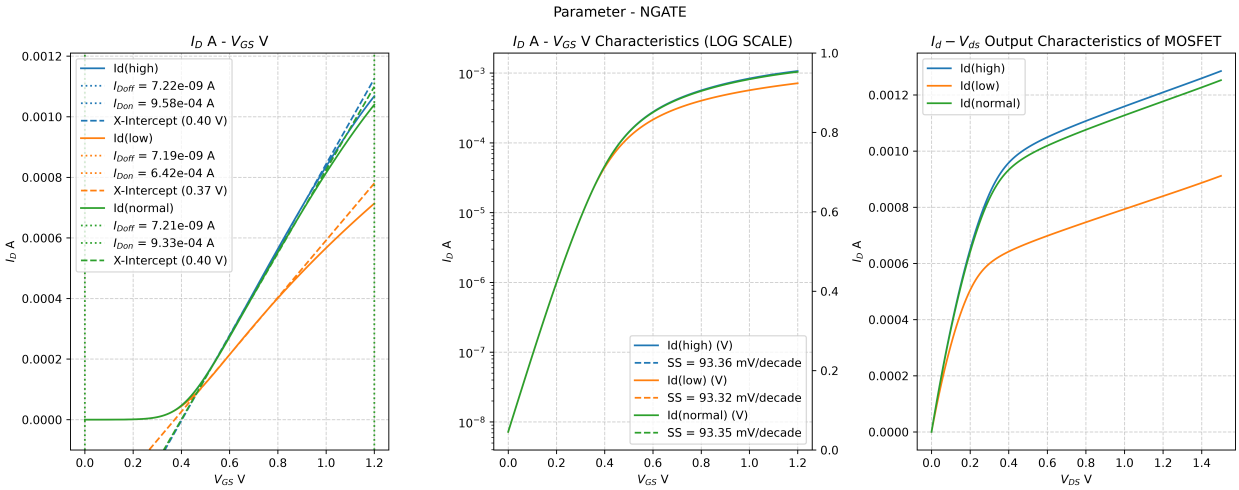


Figure 21: Exp. 11: effect of varying NGATE. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.37	93.32	642	7.19
Nominal	0.40	93.35	933	7.21
High	0.40	93.36	958	7.22

Table 17: Extracted parameters, Exp. 11.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.12 Experiment 12: U0

Values (low / nominal / high): 300 / 491 / 700 $\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$

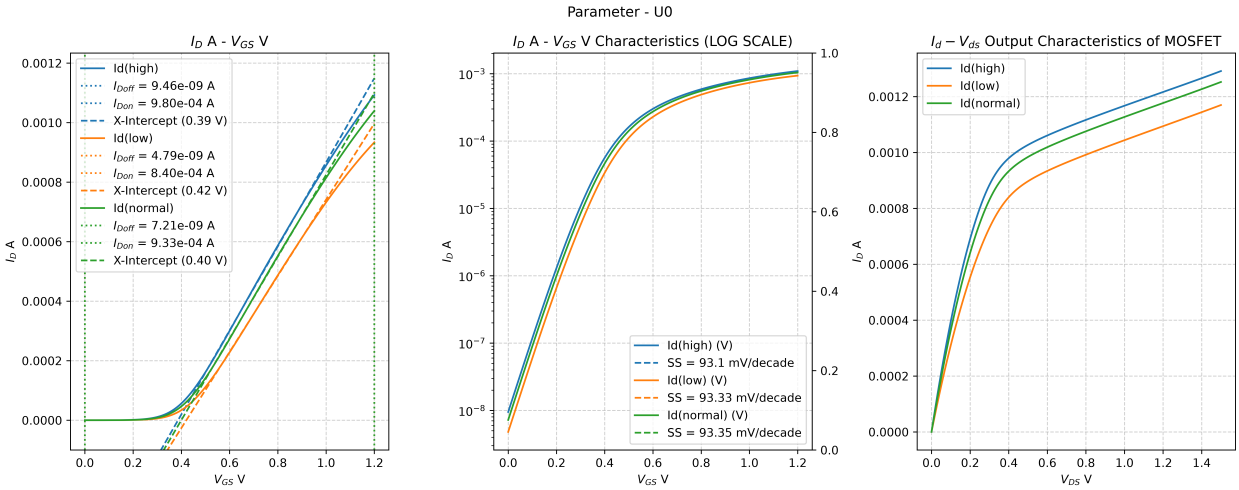


Figure 22: Exp. 12: effect of varying U0. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.42	93.33	840	4.79
Nominal	0.40	93.35	933	7.21
High	0.39	93.10	980	9.46

Table 18: Extracted parameters, Exp. 12.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.13 Experiment 13: VSAT

Values (low / nominal / high): 0.8×10^7 / 1.24×10^7 / 2×10^7 cm/s

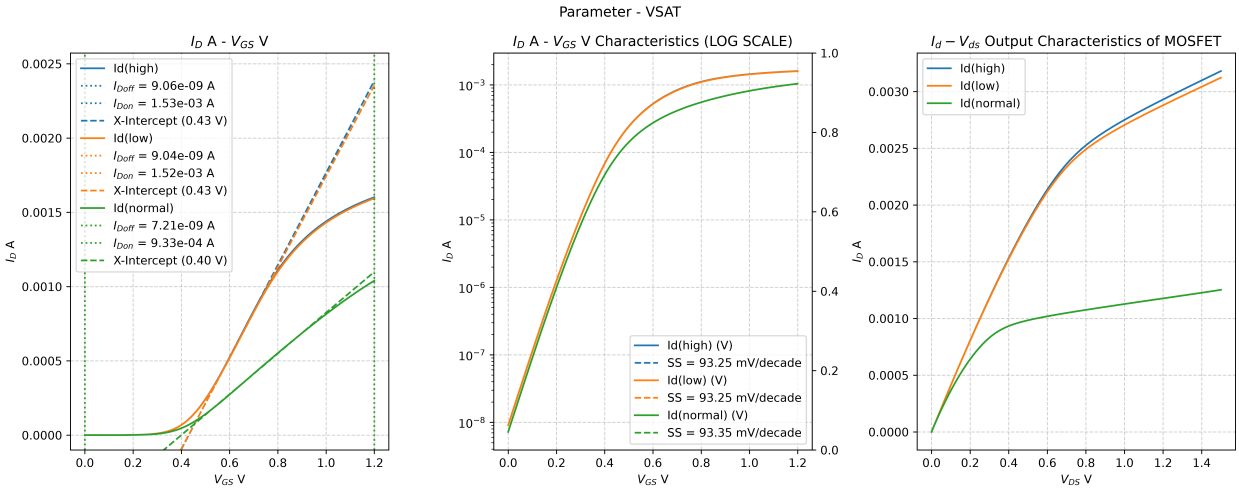


Figure 23: Exp. 13: effect of varying VSAT. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.43	93.25	1520	9.04
Nominal	0.40	93.35	933	7.21
High	0.43	93.25	1530	9.06

Table 19: Extracted parameters, Exp. 13.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.14 Experiment 14: RDSW

Values (low / nominal / high): 50 / 165 / 400 $\Omega \cdot \mu$m

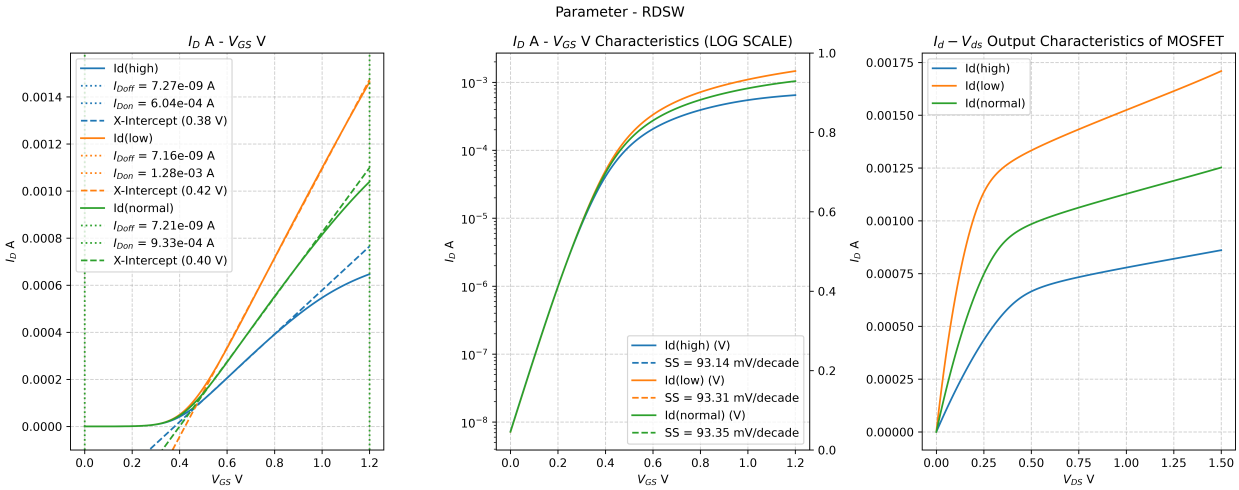


Figure 24: Exp. 14: effect of varying RDSW. Left: I_D–V_G at $V_{DS} = 0.4$ V with V_{TH} extraction; middle: same on a log scale with subthreshold swing; right: I_D–V_D at $V_{GS} = 1.1$ V.

Case	V_{TH} (V)	SS (mV/dec)	I_{ON} (μA/μm)	I_{OFF} (nA/μm)
Low	0.42	93.31	1280	7.16
Nominal	0.40	93.35	933	7.21
High	0.38	93.14	604	7.27

Table 20: Extracted parameters, Exp. 14.

Observations

[Describe the trend in V_{TH}, I_{ON}, I_{OFF}, slope/saturation.]

Device-physics explanation

[Connect the trend to device physics. If it contradicts expectation, explain how SPICE (BSIM) treats this parameter.]

5.15 Overall Summary of S5

Exp.	Parameter	V_{TH}	I_{ON}	I_{OFF}	Matches expectation? (Y/N, reason)
1	VTH0	↑	↓	↓	
2	TOXE	↑	↓	↓	
3	L	↑	↓	↓	
4	XL	↑	↓	↓	
5	LINT	↓	↑	↑	
6	W	≈	≈	≈	
7	WINT	≈	↓	↓	
8	NDEP	↑	↓	↑	
9	NSD	≈	≈	≈	
10	XJ	≈	≈	≈	
11	NGATE	≈	↑	≈	
12	U0	↓	↑	↑	
13	VSAT	≈	n.m.	≈	
14	RDSW	↓	↓	≈	

Table 21: Trend summary (↑, ↓, ≈ as the parameter increases; n.m. = non-monotonic).

Conclusion

[Which parameters dominate V_{TH}, I_{ON}, I_{OFF}? Which behave unexpectedly in SPICE and why?]

6 S6 – Approximate Match to TCAD

Circuit / Testbench

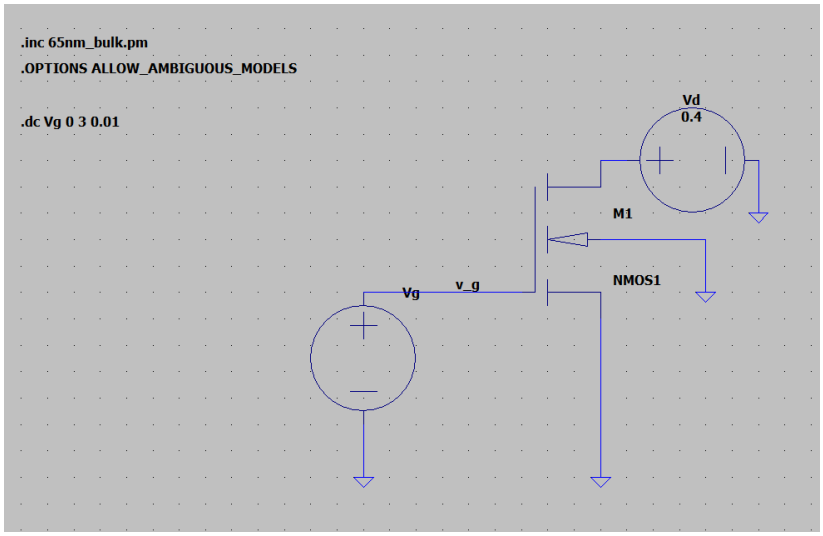
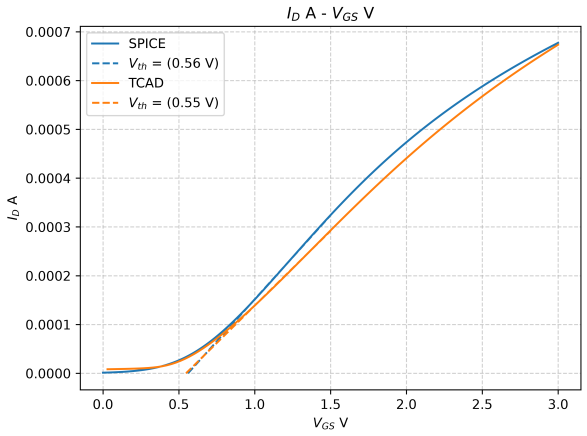


Figure 25: LTspice testbench used for the TCAD comparison.

```
.dc Vg 0 3 0.01/ .dc Vd 0 3 0.01
```



(a) I_D–V_G, $V_{DS} = 0.4$ V (linear)

Insert plot
s6-idvd-match.png

(b) I_D–V_D, $V_{GS} = 0.9$ V (linear)

Figure 27: TCAD (markers) versus tuned SPICE (lines).

Parameter	PTM value	My value	Reason for change
VTH0	0.423	0.61	To right shift the whole Id-Vg
voff	-0.13	-0.6	To pull up the low Voff value in SPICE
nfactor	1.9	5	For above process and lesser v_{th} shift and almost equal subthrehold volta
u0	0.0491	0.03	To pull the currents in saturation region
VSAT	124340	74340	same reason as above

Table 22: Parameters modified in `nfit`.

7 S7 – CMOS Inverter

7.1 Circuit and Parameters

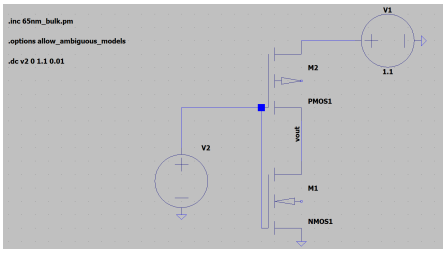


Figure 28: LTspice schematic of the CMOS inverter.

Parameter	nMOS	pMOS
L	65 nm	65 nm
W	1 μm	2 μm
V_{DD}	1.1 V	1.1 V

7.2 Voltage Transfer Characteristic

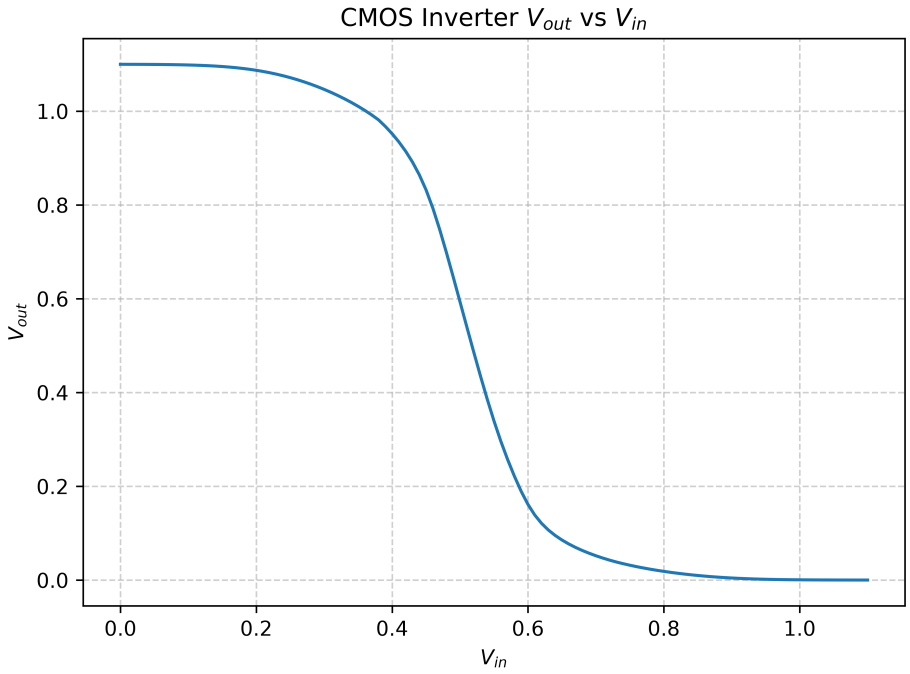


Figure 29: V_{out} versus V_{in} ($0 \to V_{DD}$); mark V_{OH}, V_{OL}, V_M and the gain region.

Quantity	Value
V_{OH} (V)	
V_{OL} (V)	
Switching threshold V_M (V)	
Max. gain $\lvert dV_{out}/dV_{in} \rvert$	

Table 23: Inverter metrics from the VTC.

Device-physics explanation

[Explain operation in each region (NMOS off/PMOS on, both in saturation, NMOS on/PMOS off) and why the circuit is called an "inverter".]

Conclusion

[Key takeaway, effect of the $W_p/W_n = 2$ ratio on V_M.]

8 S8 – GitHub Repository

- Repository name: `LTSPICE`
- URL: _____

Python script	Purpose
`s2_transfer.py`	S2: I_D–V_G plots and extraction
`s3_output.py`	S3: I_D–V_D family, g_{ds}, r_o
`s4_physics.py`	S4: model overlay
`s5_param.py`	S5: parametric sweeps
`s6_fit.py`	S6: TCAD vs SPICE overlay
`s7_inverter.py`	S7: VTC plot

Table 24: Scripts committed to the repository (rename to match yours).

9 Overall Conclusions

1. *[Conclusion 1]*
2. *[Conclusion 2]*
3. *[Conclusion 3]*
4. *[Conclusion 4]*

References

1. EC2L104 Course Handout: *A Beginner's Manual to SPICE Modelling in LTspice.*
2. PTM – Predictive Technology Model, 65 nm BSIM4 model card.
3. *[Any additional references.]*

A Appendix: Python Code Listings

```
# paste key code snippets here (full code is in the GitHub repo)
```

B Appendix: Full Model Card

```
* paste the nfit model card here
```